STEM Career Picker - REBIRTH -

A Project Write-up for a STEM Course Selector for and by PSHS Scholars
Computer Science 2 Quarter 4 Final Project

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Problem Statement

We've all had that phase where the only thing people would ask you is "What do you want to be when you grow up?" Now, as Pisay scholars, it's only upgraded to "What STEM course do you want to take?" Some of us had direct answers to both, a clear vision of their future, but others had a hard time choosing. Medicine or engineering? Technical or creative course? Lab or field work? Questions like that ringed in our head every time we were asked that question, and it remained there, with no one to help us figure out our own set goal. That is a problem. Without knowing which direction you want to go after graduation—which will fly by faster than you expect—you wouldn't be able to pick an elective for senior high school. You won't know what topic actually catches your eye in the research subject, and many, many more people will repeat the question "What STEM course do you want to take?" if these people do not figure out a course that calls to them.

Project Objectives

Objective 1

To aid Pisay students in picking a possible career path if they have not decided yet. Particularly before they reach grade 11 so they can choose which elective subjects to take based on the result of this quiz, or before graduation, basing off what electives they took in senior high school (grade 11-12).

Objective 2

To learn more about STEM careers to help even the gamemakers decide what course they can take.

For instance, one of the results required in this quiz is a physicist. We would have to research what a physicist's usual interests and required subjects are, therefore learning more about them, and if one gamemaker aligns their interests and subjects, then they can have the opportunity of being a physicist open to them.

Planned Features

Feature 1

Explanation presented to the user about the purpose of this STEM course selector (see **Project Objectives**)

Example:

> Welcome to STEM Career Picker: Rebirth!
>
> This game will try its best to determine a stable STEM job for you based on your experiences and opinions on things.
>
> The gist of this fun test is that you'll be answering multiple questions with either the first or second prompt
>
> using 1 and 2 as responses

Feature 2

Alongside the main result (a STEM Course), there will be a list of possible electives and specific jobs assigned to the result.

Example:

the STEM aligned course you could have been in your premature mortal life would be…
Biology

> You could have learned so much...
> Perhaps through these electives:
> Grade 11: ["Biology Science Specialization (Level 1 or 2)", "Research 1 (Focus on issues related to biology)
> Grade 12: ["Biology Science Specialization (Level 1 or 2)", "Research 2 (Focus on issues related to biology)"
> You could have given back to society so much…
> There were opportunities a plenty:
> Careers: ["General Biologist", "Pathologist", "Biomedical Engineer", "Research Scientist (Biology Specialty)", "Field Biologist", "Environmental Engineer", "Educator for Biology-Related Subjects (Also Requires Education Degree)"]
> It certainly is a wither...
> Your mortal self truly was a special one...
> So many possibilities...wasted

Feature 3

Our game will follow through with not just a STEM course selector, but also a storyline focusing on the afterlife.
The user will be granted a chance to get information before entering purgatory
To find out which STEM course suits them best, they will have to choose “1 - Enlightenment” in a list of options presented to them by a being in the afterlife
To reminisce their past life, they will choose “2 - Reminisce”
To go straight to an ending, they will choose “3 - Board”

Planned Inputs and Outputs

Inputs

- The user will be asked to input their name
- In the beginning, the user is in the afterlife. They will be asked to input a number from 1 - 3. This will decide if they wish to be enlightened, reminisce, or board the boat to the “forever” afterlife
- If the user chooses option 1:
    - The user will be asked to input either 1 or 2 depending on their thoughts on a presented question about STEM
- If the user chooses option 2:
    - The user will be asked to input their choice of what knowledge to find out:
        - If the user chooses choice 1:
            - They will be asked to input their new name
        - If the user chooses choice 2:
            - No more inputs required
        - If the user chooses choice 3:
            - They will be asked to pick a new origin story
- If the user chooses option 3:
    - The user will be asked to input their answer to one question related to science that is presented to them.
- If the user input anything not 1-3:
    - Error message

Outputs

- ASCII art of Title in Main Menu
- Main Menu + Instructions
- Storytelling portion about how the user ended up in the afterlife
- Outputs asking the user to input something (e.g., name, choice)
- Option 1:

    - The series of questions that will determine what STEM course/classes/careers is best suited for the user.
    - The end result that will print said course/classes/careers best suited for the user.
    - If the user inputs anything other than 1 or 2:
        - Error message
- Option 2:
    - If the user chooses choice 1:
        - An output asking the user to change their name
    - If the user chooses choice 2:
        - The output will be a judgemental comment from the person the user is speaking to.
    - If the user chooses choice 3:
        - Output beckoning the user to pick their new origin story
        - Output of a comment from the person about the origin story the user picked
    - If the user inputs anything other than 1-3
        - Error message
- Option 3:
    - Storytelling portion
    - The question that will determine the fate of the user
      "Is science an infinite source of knowledge, a life's journey, a mere tool for phenomena, or life itself?"
      1. Infinite source knowledge
      2. Life's journey
      3. Tool for phenomena
      4. Understanding of the world
      5. No answer
    - If the user chooses options 1-4:
        - Dialogue from the person saying that the user will be able to get an eternal pursuit for knowledge
    - If the user chooses option 5:
        - Dialogue from the person saying that the user will be stuck without the pursuit of any other knowledge
    - If the user inputs anything other than 1-5:
        - Dialogue from the person saying that the user will be stuck in eternal wonder with no answers

Logic Plans

Pseudocode

Main Program

```
Display ASCII Art

Display Introduction and Narration

Display Options
   1 = Enlightenment
   2 = Reminiscence
   3 = Board

Input choice

If choice == 1:
  Call game()

Elif choice == 2:
   Call background(name)

Elif choice == 3:
   Call end()

Else:
   Display error message
```

game()

```
Display introduction text

Ask: Do you enjoy problem solving? (1 = Yes, 2 = No)

If Yes:
   Ask: Creating or Thinking?

   If Creating:
      Ask: Coding or Building?

      If Coding:
         Display course[0]

      Elif Building:
         Ask: Specific or General?

         If Specific:
```

```
            Display course[1]
        Elif General:
            Display course[2]
        Else:
            Display error message

    Elif Thinking:
        Ask: Nature or Numbers?

        If Nature:
            Ask: Open field or Laboratory?

            If Open:
                Display course[3]
            Elif Lab:
                Display course[6]
            Else:
                Display error message

        Elif Numbers:
            Display error text
            Display ellipsis
            Ask: Familiarization or Memorization?

            If Familiarization:
                Ask: Identify structures or Predict?

                If Identify:
                    Display course[7]
                Elif Predict:
                    Display course[4]
                Else:
                    Display error message

            Elif Memorization:
                Display course[5]
            Else:
                Display error message

        Else:
            Display error message

    Elif:
        Display error message

Elif No:
    Ask: Application or Derivation?

    If Application:
        Ask: Digital or Outside work?

        If Digital:
```

```
        Ask: Quality or Performance?

        If Quality:
            Display course[8]
        Elif Performance:
            Display course[9]
        Else:
            Display error message

    Elif Outside:
        Ask: Large-scale or High-value?

        If Large-scale:
            Display course[10]
        Elif High-value:
            Display course[11]
        Else:
            Display error message

    Else:
        Display error message

  Elif Derivation:
    Ask: Theoretical or Applied?

    If Theoretical:
        Display course[12]
    Elif Applied:
        Display course[13]
    Else:
        Display error message

  Else:
    Display error message

Else:
  Display error message
```

background(name)

```
Display Narration

Display Options
  1 = wake of Dawn - Name Change
  2 = bask of Noon - Ending up in Afterlife
  3 = last of Dusk - Origin Story

Input ray

If ray == 1:
  Ask: Do you wish to change your name? (1 if yes, 2 if no)
```

```
    If yes:
        Input name
        Display new name

Elif ray == 2:
    Display lore about reputation and records

Elif ray == 3:
    Ask: Do you want to change your origin story?
    If yes:
        Display origin options
            Option 1 = Commuting to work
            Option 2 = Taking a walk along the avenue
            Option 3 = Jog through the bus stop
            Option ■ = Opportunity to ■■■■■■ by the ■■■■■■■.

        Input Origin Story

        If origin_story == 1:
            Display commuting story + commentary

        Elif origin_story == 2:
            Display walking story + commentary

        Elif origin_story == 3:
            Display jogging story + commentary

        Elif origin_story == 4:
            Display error

        Else:
            Display error message
```

end()

```
Display Narration

Ask Philosophical Question

Display Options

    1 = Infinite source knowledge
    2 = Life's journey
    3 = Tool for phenomena
    4 = Understanding of the world
    5 = No answer

Input final_fate

If final fate == 1 or final fate == 2 or final fate == 3 or final fate == 4:
    Display good ending - eternal pursuit of knowledge
```

```
Elif final_fate == 5:
   Display bad ending 1 - Banished from pursuing knowledge

Else:
   Display bad ending 2 - Soul is abandoned
```

References

Section III: Subject Matrix (Revised Version). (2025, May 14). https://pshs.edu.ph/wp-content/uploads/2025/06/PSHS-6-Year-Subject-Matrix_May-14-2025.pdf